VolunTier

DIGT2107: Practice of Software Development
Project Iteration 3: Testing and Initial Development
Course: DIGT2107 - Fall Term 2026. Instructor: Dr. May Haidar. Due date: November 8, 2026.

1. Introduction

Project Name: VolunTier
Team Number: 3
Team Members: Jana Hussien (222143515), Mohammad Jalili (221243084), Anwar Bonunagui (222290266), Crystal Mumtaz-Shah (220409777)

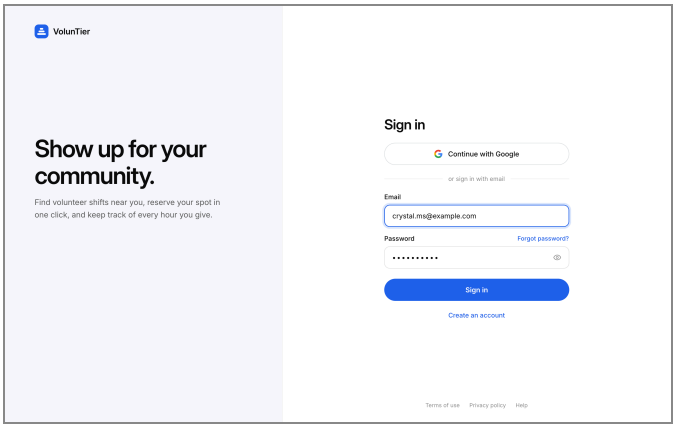
Document Overview: This document covers Iteration 3 of VolunTier, a Java desktop application that helps community organizations post volunteer opportunities, register volunteers, record attendance and hours, and send notifications. It shows how the test cases link to the functional and non-functional requirements from Iteration 2. It also gives the repository strategy, the task allocation, and the sprint plan.

2. Iteration Goals and Objectives

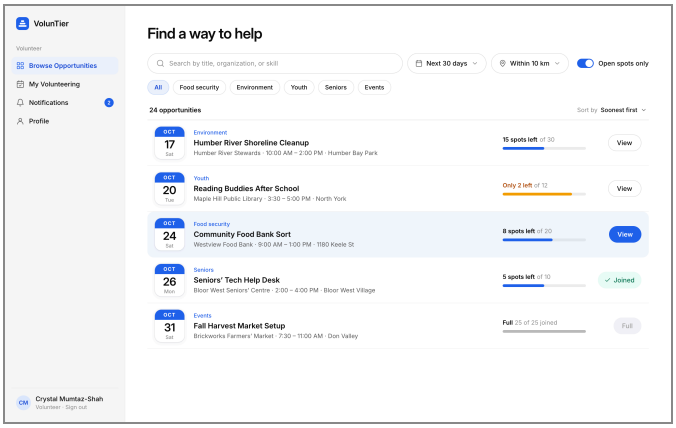
- Implement core functionality for the high-priority user stories: Role-Based Login (US1), Browse and Search Opportunities (US2), Register for an Opportunity (US3), and Manage Opportunities (US6).
- Write and run unit tests that map to specific requirements. Each test uses a test case ID in its name.
- Deliver an initial prototype that shows the link between test cases and requirements.

Prototype Overview

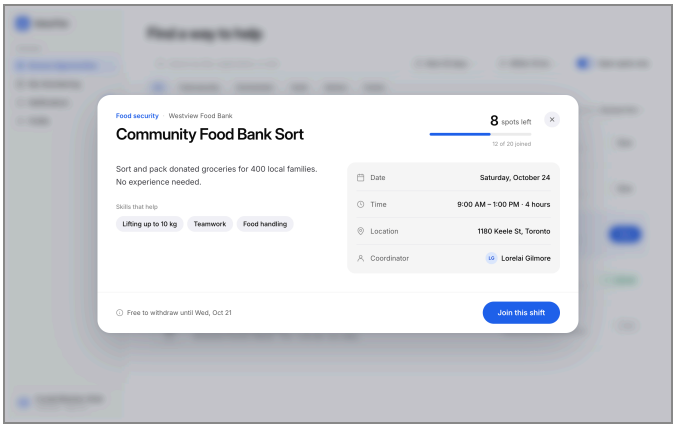
The Iteration 2 prototype has seven wireframes with mock data, and one flow diagram. The Iteration 3 build replaces mock data with a working Java application, connected screens, and data storage.



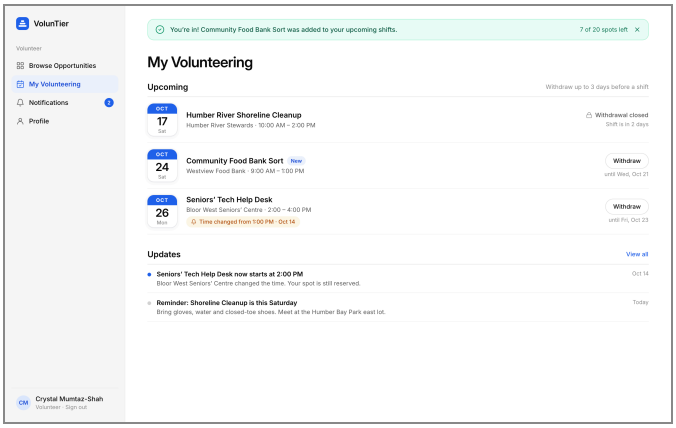
Sign In (FR-001)



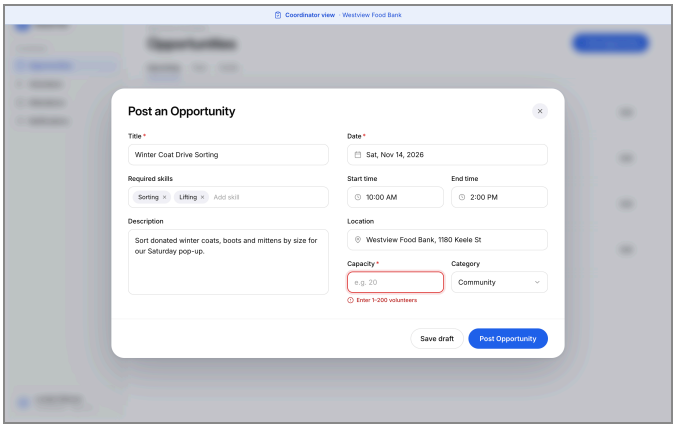
Browse Opportunities (FR-002)



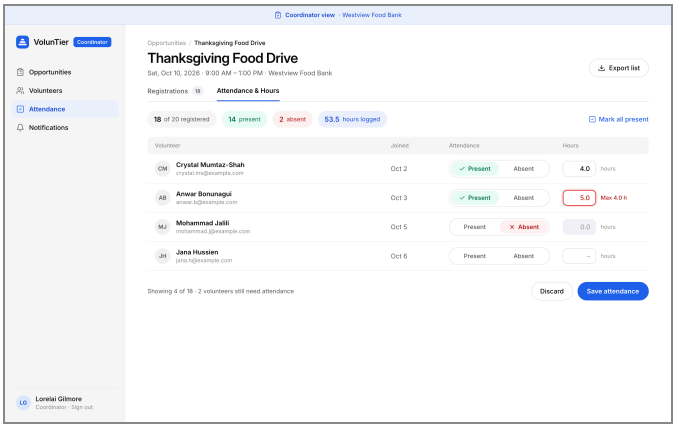
Opportunity Details and Register (FR-003)



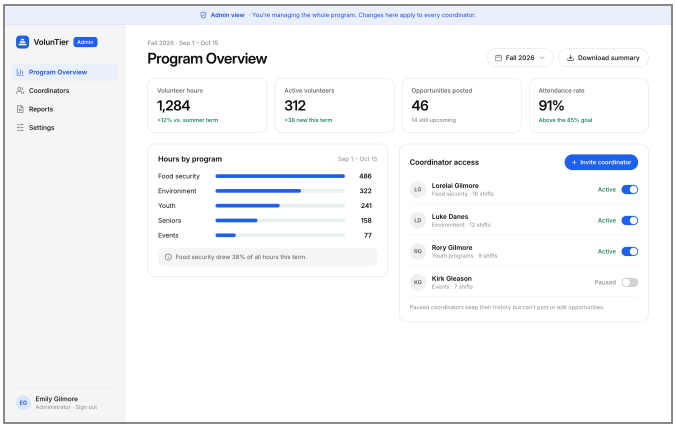
My Volunteering (FR-004, FR-005)



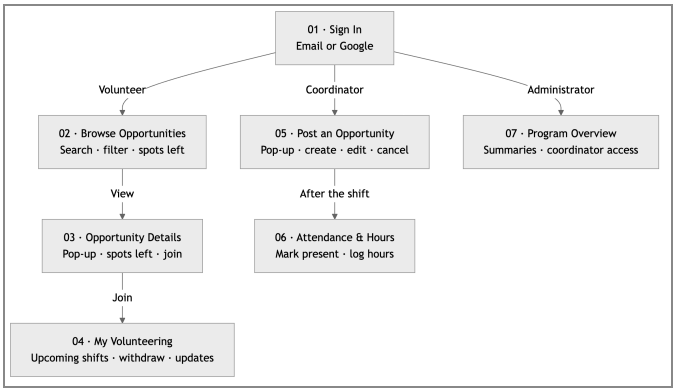
Post an Opportunity (FR-006)



Attendance and Hours (FR-007)



Program Overview (FR-008)



Screen flow

3. Updated Requirements and Use Cases

3.1 Functional Requirements

ID	Requirement	User Story	Priority
FR-001	The system shall allow users to log in and access features based on their assigned role: volunteer, coordinator, or administrator.	US1	High
FR-002	The system shall allow volunteers to browse, search, and filter available volunteer opportunities.	US2	High
FR-003	The system shall allow volunteers to register for available opportunities while preventing duplicate registrations and registrations beyond the capacity limit.	US3	High
FR-004	The system shall allow volunteers to withdraw from an opportunity more than three days before the event's scheduled start time.	US4	Medium
FR-005	The system shall allow volunteers to view their upcoming registrations, participation history, and completed volunteer hours.	US5	Medium
FR-006	The system shall allow volunteer coordinators to create, post, edit, and cancel volunteer opportunities.	US6	High
FR-007	The system shall allow volunteer coordinators to view registrations and record volunteer attendance and completed hours.	US7	Medium
FR-008	The system shall allow organization administrators to manage coordinator access and review program summaries.	US8	Medium
FR-009	The system shall send registered volunteers an in-app or email notification when an opportunity's date, time, location, or status changes.	US9	Medium

IDs FR-001 to FR-009 match requirements 1 to 9 in the Iteration 2 document.

3.2 Non-Functional Requirements

ID	Quality	Requirement
NFR-001	Performance	Opportunity lists, registration results, and participation records appear within two seconds under normal use.
NFR-002	Usability	The Java desktop interface uses clear labels, readable text, consistent navigation, and understandable error messages.
NFR-003	Reliability	The system saves registration, withdrawal, attendance, and volunteer-hour records accurately without duplicate records.
NFR-004	Security	The system restricts features by user role (volunteer, coordinator, administrator).
NFR-005	Scalability	The database supports hundreds of users and thousands of participation records.
NFR-006	Validation	The system validates required fields, dates, capacity limits, and volunteer hours before saving.
NFR-007	Maintainability	The application uses a modular structure. Interface, business rules, and data storage can change separately.

3.3 Use Cases

ID	Use Case	Actor	Requirement
UC-01	Browse Available Opportunities	Volunteer	FR-002
UC-02	Search and Filter Opportunities	Volunteer	FR-002
UC-03	Register for an Opportunity	Volunteer	FR-003
UC-04	Withdraw from an Opportunity	Volunteer	FR-004
UC-05	View Participation History	Volunteer	FR-005
UC-06	Manage and Post Opportunities	Coordinator	FR-006, FR-009

UC-07	View Volunteer Registrations	Coordinator	FR-007
UC-08	Record Attendance and Hours	Coordinator	FR-007
UC-09	Manage Coordinator Access	Administrator	FR-008
UC-10	Review Program Summaries	Administrator	FR-008

The full use case descriptions (main flow, alternative flow, postconditions) are in the Iteration 2 document. No use case changed in Iteration 3.

4. Requirement to Test Case Traceability

4.1 Traceability Matrix: Functional Requirements

Req. ID	Requirement Description	Test Case ID	Test Case Description	Coverage
FR-001	The system shall allow users to log in and access features based on their assigned role: volunteer, coordinator, or administrator.	TC-001, TC-002, TC-003, TC-004, TC-005, TC-006, TC-007	TC-001: Login - Valid Volunteer Credentials TC-002: Login - Valid Coordinator Credentials TC-003: Login - Valid Administrator Credentials TC-004: Login - Invalid Password TC-005: Login - Empty Required Fields TC-006: Role Restriction - Volunteer Cannot Access Coordinator Features TC-007: Role Restriction - Coordinator Cannot Access Administrator Features	Fully Covered
FR-002	The system shall allow volunteers to browse, search, and filter available volunteer opportunities.	TC-008, TC-009, TC-010, TC-011, TC-012	TC-008: Browse Opportunities - List Displays Details TC-009: Search Opportunities - Matching Term TC-010: Search Opportunities - No Matches TC-011: Filter Opportunities - By Category or Date TC-012: View Opportunity Details	Fully Covered
FR-003	The system shall allow volunteers to register for available opportunities while preventing duplicate registrations and registrations beyond the capacity limit.	TC-013, TC-014, TC-015, TC-016	TC-013: Register for Opportunity - Space Available TC-014: Register for Opportunity - Duplicate Registration TC-015: Register for Opportunity - Capacity Reached TC-016: Register for Opportunity - Last Remaining Space	Fully Covered
FR-004	The system shall allow volunteers to withdraw from an opportunity more than three days before the event's scheduled start time.	TC-017, TC-018, TC-019	TC-017: Withdraw - More Than Three Days Before Event TC-018: Withdraw - Within Three Days of Event TC-019: Withdraw - Exactly Three Days Boundary	Fully Covered
FR-005	The system shall allow volunteers to view their upcoming registrations, participation history, and completed volunteer hours.	TC-020, TC-021, TC-022	TC-020: My Volunteering - Upcoming and History TC-021: My Volunteering - Hours Show After Coordinator Records Them TC-022: My Volunteering - No Records	Fully Covered

FR-006	The system shall allow volunteer coordinators to create, post, edit, and cancel volunteer opportunities.	TC-023, TC-024, TC-025, TC-026, TC-027	TC-023: Create Opportunity - Valid Inputs TC-024: Create Opportunity - Missing Required Fields TC-025: Create Opportunity - Invalid Date or Capacity TC-026: Edit Opportunity - Valid Change TC-027: Cancel Opportunity - Registered Volunteers Notified	Partially Covered
FR-007	The system shall allow volunteer coordinators to view registrations and record volunteer attendance and completed hours.	TC-021, TC-028, TC-029, TC-030	TC-021: My Volunteering - Hours Show After Coordinator Records Them TC-028: View Registrations for an Opportunity TC-029: Record Attendance and Hours - Valid TC-030: Record Hours - Invalid Hours	Partially Covered
FR-008	The system shall allow organization administrators to manage coordinator access and review program summaries.	TC-031, TC-032, TC-033, TC-034	TC-031: Grant Coordinator Access TC-032: Remove Coordinator Access TC-033: Review Program Summaries TC-034: Program Summaries - No Data For Period	Partially Covered
FR-009	The system shall send registered volunteers an in-app or email notification when an opportunity's date, time, location, or status changes.	TC-027, TC-035, TC-036	TC-027: Cancel Opportunity - Registered Volunteers Notified TC-035: Notification - Opportunity Time or Location Changed TC-036: Notification - Unregistered Volunteer Not Notified	Partially Covered

Coverage shows the planned coverage from test design. The team must confirm it after the tests run.

4.2 Traceability Matrix: Non-Functional Requirements

Req. ID	Requirement Description	Test Case ID	Test Case Description
NFR-001	Performance: Opportunity lists, registration results, and participation records appear within two seconds under normal use.	TC-037	TC-037: Performance - Lists Load Within Two Seconds
NFR-002	Usability: The Java desktop interface uses clear labels, readable text, consistent navigation, and understandable error messages.	None	No test case yet
NFR-003	Reliability: The system saves registration, withdrawal, attendance, and volunteer-hour records accurately without duplicate records.	TC-038	TC-038: Reliability - No Duplicate Records On Rapid Double Click
NFR-004	Security: The system restricts features by user role (volunteer, coordinator, administrator).	TC-006, TC-007	TC-006: Role Restriction - Volunteer Cannot Access Coordinator Features TC-007: Role Restriction - Coordinator Cannot Access Administrator Features
NFR-005	Scalability: The database supports hundreds of users and thousands of participation records.	None	No test case yet

NFR-006	Validation: The system validates required fields, dates, capacity limits, and volunteer hours before saving.	TC-005, TC-024, TC-025, TC-030	TC-005: Login - Empty Required Fields TC-024: Create Opportunity - Missing Required Fields TC-025: Create Opportunity - Invalid Date or Capacity TC-030: Record Hours - Invalid Hours
NFR-007	Maintainability: The application uses a modular structure. Interface, business rules, and data storage can change separately.	None	No test case yet

Coverage status is not required for non-functional requirements.

4.3 Explanation

How the test cases validate each requirement. Each functional requirement has at least two test cases. Each has one test for the normal path and one test for a rejected input or a blocked action. Registration (FR-003) and withdrawal (FR-004) also have boundary tests for the last free space and the exact three-day cutoff.

Gaps.

- FR-006: Add tests for invalid input on edit, and for a coordinator who edits an opportunity that another coordinator owns.
- FR-007: Add tests for attendance on an event that has not yet taken place, and for hours above the event length.
- FR-008: Add a test that the summary totals equal the stored records.
- FR-009: Email delivery is untested. Add a test for each changed field (date, time, location, status).
- NFR-002 (Usability), NFR-005 (Scalability), and NFR-007 (Maintainability) have no test case. Usability is checked by team review of the screens. Scalability needs a data load script. Maintainability is checked by code review of the module structure.

Coverage status. 5 functional requirements are fully covered. 4 are partially covered. None are uncovered. The team will update this status after the tests run.

5. Test Plan and Results

5.1 Test Plan

- **Scope:** all functional requirements FR-001 to FR-009. Priority goes to FR-001, FR-002, FR-003, and FR-006 (high-priority stories).
- **Unit tests:** JUnit 5 tests for business rules. These rules are role checks, duplicate and capacity checks, the three-day withdrawal rule, and input validation. Test names include the test case ID, for example `TC014_duplicateRegistrationIsRejected`.
- **Integration tests:** tests of the data storage layer for registration, withdrawal, attendance, and hours records. These tests check that no duplicate record is created.
- **Manual tests:** the team runs the screen-level test cases in section 6 on the Java interface.
- **Environment:** Java desktop application, JUnit 5, test database with sample data (hundreds of users, thousands of records for the performance test).
- **Entry criteria:** the feature is merged to the `development` branch and builds without error.
- **Exit criteria:** all High priority test cases pass. No High priority defect stays open.
- **Risk:** the team can run out of time for the notification feature (FR-009). The sprint plan delays this feature to Week 2 so it does not block the high-priority stories.

5.2 Results

The team has 38 test cases (15 High, 19 Medium, 4 Low). The team has not run them yet. The Actual Result and Status columns in section 6 stay empty until the test run. The team will add the results to this table after Week 2.

Priority	Test cases	Run	Pass	Fail	Blocked
High	15	0	-	-	-
Medium	19	0	-	-	-
Low	4	0	-	-	-
Total	38	0	-	-	-

6. Detailed Test Case Descriptions

TC-001

Test Case ID	TC-001
Related Requirement(s)	FR-001
Title	Login - Valid Volunteer Credentials
Preconditions	A volunteer account exists. The user is on the Sign In screen.
Test Steps	1. Enter the volunteer email and password 2. Click "Sign In"
Expected Result	The volunteer dashboard opens. Only volunteer features show.
Actual Result	(To be filled after test execution)
Status	(Pass/Fail/Blocked)
Priority	High

TC-002

Test Case ID	TC-002
Related Requirement(s)	FR-001
Title	Login - Valid Coordinator Credentials
Preconditions	A coordinator account exists. The user is on the Sign In screen.
Test Steps	1. Enter the coordinator email and password 2. Click "Sign In"
Expected Result	The coordinator dashboard opens with opportunity management features.
Actual Result	(To be filled after test execution)
Status	(Pass/Fail/Blocked)
Priority	High

TC-003

Test Case ID	TC-003
Related Requirement(s)	FR-001
Title	Login - Valid Administrator Credentials
Preconditions	An administrator account exists. The user is on the Sign In screen.
Test Steps	1. Enter the administrator email and password 2. Click "Sign In"
Expected Result	The administrator dashboard opens with user management and program summaries.
Actual Result	(To be filled after test execution)
Status	(Pass/Fail/Blocked)
Priority	High

TC-004

Test Case ID	TC-004
Related Requirement(s)	FR-001
Title	Login - Invalid Password
Preconditions	A volunteer account exists. The user is on the Sign In screen.
Test Steps	1. Enter a valid email and a wrong password 2. Click "Sign In"
Expected Result	Login fails. A clear error message shows. The user stays on the Sign In screen.
Actual Result	(To be filled after test execution)
Status	(Pass/Fail/Blocked)
Priority	High

TC-005

Test Case ID	TC-005
Related Requirement(s)	FR-001, NFR-006
Title	Login - Empty Required Fields
Preconditions	The user is on the Sign In screen.
Test Steps	1. Leave email and password empty 2. Click "Sign In"
Expected Result	The system marks the required fields and shows an error. No login occurs.
Actual Result	(To be filled after test execution)
Status	(Pass/Fail/Blocked)
Priority	Medium

TC-006

Test Case ID	TC-006
Related Requirement(s)	FR-001, NFR-004
Title	Role Restriction - Volunteer Cannot Access Coordinator Features
Preconditions	The user is logged in as a Volunteer.
Test Steps	1. Look for "Post an Opportunity" and "Attendance & Hours" in the navigation 2. Try to open the coordinator page directly (URL or menu)
Expected Result	The features are hidden or access is denied. No coordinator data shows.
Actual Result	(To be filled after test execution)
Status	(Pass/Fail/Blocked)
Priority	High

TC-007

Test Case ID	TC-007
Related Requirement(s)	FR-001, NFR-004
Title	Role Restriction - Coordinator Cannot Access Administrator Features
Preconditions	The user is logged in as a Coordinator.
Test Steps	1. Look for "Manage Coordinator Access" and "Program Overview" in the navigation 2. Try to open the administrator page directly
Expected Result	Access is denied. No administrator data shows.
Actual Result	(To be filled after test execution)
Status	(Pass/Fail/Blocked)
Priority	High

TC-008

Test Case ID	TC-008
Related Requirement(s)	FR-002
Title	Browse Opportunities - List Displays Details
Preconditions	The user is logged in as a Volunteer. At least three opportunities exist.
Test Steps	1. Open the Browse Opportunities page
Expected Result	Each opportunity shows date, location, description, and remaining spaces.
Actual Result	(To be filled after test execution)
Status	(Pass/Fail/Blocked)
Priority	High

TC-009

Test Case ID	TC-009
Related Requirement(s)	FR-002
Title	Search Opportunities - Matching Term
Preconditions	The user is logged in as a Volunteer. An opportunity titled "Food Bank Sorting" exists.
Test Steps	1. Open Browse Opportunities 2. Enter "Food Bank" in the search field
Expected Result	Only matching opportunities show, including "Food Bank Sorting".
Actual Result	(To be filled after test execution)
Status	(Pass/Fail/Blocked)
Priority	High

TC-010

Test Case ID	TC-010
Related Requirement(s)	FR-002
Title	Search Opportunities - No Matches
Preconditions	The user is logged in as a Volunteer.
Test Steps	1. Open Browse Opportunities 2. Enter a term that matches nothing (e.g., "zzzzqq")
Expected Result	A "no matches" message shows. The user can change the search or filters.
Actual Result	(To be filled after test execution)
Status	(Pass/Fail/Blocked)
Priority	Medium

TC-011

Test Case ID	TC-011
Related Requirement(s)	FR-002
Title	Filter Opportunities - By Category or Date
Preconditions	The user is logged in as a Volunteer. Opportunities exist in more than one category or date.
Test Steps	1. Open Browse Opportunities 2. Select a filter value
Expected Result	The list shows only opportunities that match the filter.
Actual Result	(To be filled after test execution)
Status	(Pass/Fail/Blocked)
Priority	Medium

TC-012

Test Case ID	TC-012
Related Requirement(s)	FR-002
Title	View Opportunity Details
Preconditions	The user is logged in as a Volunteer. Opportunities exist.
Test Steps	1. Open Browse Opportunities 2. Select one opportunity
Expected Result	The Opportunity Details page shows title, description, date, time, location, capacity, and a Register button.
Actual Result	(To be filled after test execution)
Status	(Pass/Fail/Blocked)
Priority	Medium

TC-013

Test Case ID	TC-013
Related Requirement(s)	FR-003
Title	Register for Opportunity - Space Available
Preconditions	Logged in as a Volunteer. The opportunity has open spaces. The volunteer is not registered.
Test Steps	1. Open the opportunity details 2. Click "Register"
Expected Result	A confirmation shows. The opportunity appears in Upcoming Registrations. Remaining spaces fall by one.
Actual Result	(To be filled after test execution)
Status	(Pass/Fail/Blocked)
Priority	High

TC-014

Test Case ID	TC-014
Related Requirement(s)	FR-003
Title	Register for Opportunity - Duplicate Registration
Preconditions	Logged in as a Volunteer. The volunteer is already registered for the opportunity.
Test Steps	1. Open the same opportunity details 2. Click "Register"
Expected Result	The system refuses and explains that the volunteer is already registered. No second record is created.
Actual Result	(To be filled after test execution)
Status	(Pass/Fail/Blocked)
Priority	High

TC-015

Test Case ID	TC-015
Related Requirement(s)	FR-003
Title	Register for Opportunity - Capacity Reached
Preconditions	Logged in as a Volunteer. The opportunity has 0 remaining spaces.
Test Steps	1. Open the opportunity details 2. Click "Register"
Expected Result	The system refuses and explains that the opportunity is full. No registration is saved.
Actual Result	(To be filled after test execution)
Status	(Pass/Fail/Blocked)
Priority	High

TC-016

Test Case ID	TC-016
Related Requirement(s)	FR-003
Title	Register for Opportunity - Last Remaining Space
Preconditions	Logged in as a Volunteer. The opportunity has exactly 1 remaining space.
Test Steps	1. Register for the opportunity 2. Log in as a second volunteer and try to register
Expected Result	The first volunteer registers. Remaining spaces show 0. The second volunteer is refused.
Actual Result	(To be filled after test execution)
Status	(Pass/Fail/Blocked)
Priority	Medium

TC-017

Test Case ID	TC-017
Related Requirement(s)	FR-004
Title	Withdraw - More Than Three Days Before Event
Preconditions	Logged in as a Volunteer. The volunteer is registered for an event 7 days away.
Test Steps	1. Open My Volunteering 2. Select the opportunity and click "Withdraw" 3. Confirm
Expected Result	The registration is removed. A confirmation shows. The space is available again.
Actual Result	(To be filled after test execution)
Status	(Pass/Fail/Blocked)
Priority	Medium

TC-018

Test Case ID	TC-018
Related Requirement(s)	FR-004
Title	Withdraw - Within Three Days of Event
Preconditions	Logged in as a Volunteer. The volunteer is registered for an event 2 days away.
Test Steps	1. Open My Volunteering 2. Select the opportunity and click "Withdraw"
Expected Result	Withdrawal is blocked. The message explains the three-day cutoff. The registration stays.
Actual Result	(To be filled after test execution)
Status	(Pass/Fail/Blocked)
Priority	Medium

TC-019

Test Case ID	TC-019
Related Requirement(s)	FR-004
Title	Withdraw - Exactly Three Days Boundary
Preconditions	Logged in as a Volunteer. The volunteer is registered for an event exactly 72 hours away.
Test Steps	1. Open My Volunteering 2. Select the opportunity and click "Withdraw"
Expected Result	Withdrawal is blocked because the event is not more than three days away.
Actual Result	(To be filled after test execution)
Status	(Pass/Fail/Blocked)
Priority	Low

TC-020

Test Case ID	TC-020
Related Requirement(s)	FR-005
Title	My Volunteering - Upcoming and History
Preconditions	Logged in as a Volunteer. The volunteer has upcoming and past registrations.
Test Steps	1. Open My Volunteering
Expected Result	Upcoming registrations and past participation show in separate, clear sections.
Actual Result	(To be filled after test execution)
Status	(Pass/Fail/Blocked)
Priority	Medium

TC-021

Test Case ID	TC-021
Related Requirement(s)	FR-005, FR-007
Title	My Volunteering - Hours Show After Coordinator Records Them
Preconditions	A coordinator recorded 3 hours for the volunteer. Logged in as that Volunteer.
Test Steps	1. Open My Volunteering 2. View participation history
Expected Result	The history shows the opportunity as attended with 3 completed hours. The total hours include 3.
Actual Result	(To be filled after test execution)
Status	(Pass/Fail/Blocked)
Priority	Medium

TC-022

Test Case ID	TC-022
Related Requirement(s)	FR-005
Title	My Volunteering - No Records
Preconditions	Logged in as a new Volunteer with no registrations.
Test Steps	1. Open My Volunteering
Expected Result	A helpful message shows that there are no records.
Actual Result	(To be filled after test execution)
Status	(Pass/Fail/Blocked)
Priority	Low

TC-023

Test Case ID	TC-023
Related Requirement(s)	FR-006
Title	Create Opportunity - Valid Inputs
Preconditions	Logged in as a Coordinator.
Test Steps	1. Open Post an Opportunity 2. Enter title, description, future date, time, location, and capacity 10 3. Click "Post"
Expected Result	The opportunity saves. It appears in the list that volunteers see.
Actual Result	(To be filled after test execution)
Status	(Pass/Fail/Blocked)
Priority	High

TC-024

Test Case ID	TC-024
Related Requirement(s)	FR-006, NFR-006
Title	Create Opportunity - Missing Required Fields
Preconditions	Logged in as a Coordinator.
Test Steps	1. Open Post an Opportunity 2. Leave the title empty and fill other fields 3. Click "Post"
Expected Result	The system explains the missing field. Nothing is saved.
Actual Result	(To be filled after test execution)
Status	(Pass/Fail/Blocked)
Priority	High

TC-025

Test Case ID	TC-025
Related Requirement(s)	FR-006, NFR-006
Title	Create Opportunity - Invalid Date or Capacity
Preconditions	Logged in as a Coordinator.
Test Steps	1. Open Post an Opportunity 2. Enter a past date or a capacity of 0 or a negative number 3. Click "Post"
Expected Result	The system rejects the input with a clear message. Nothing is saved.
Actual Result	(To be filled after test execution)
Status	(Pass/Fail/Blocked)
Priority	Medium

TC-026

Test Case ID	TC-026
Related Requirement(s)	FR-006
Title	Edit Opportunity - Valid Change
Preconditions	Logged in as a Coordinator. An opportunity exists.
Test Steps	1. Open the opportunity for editing 2. Change the description 3. Click "Save"
Expected Result	The change saves and shows in the opportunity list.
Actual Result	(To be filled after test execution)
Status	(Pass/Fail/Blocked)
Priority	Medium

TC-027

Test Case ID	TC-027
Related Requirement(s)	FR-006, FR-009
Title	Cancel Opportunity - Registered Volunteers Notified
Preconditions	Logged in as a Coordinator. An opportunity has two registered volunteers.
Test Steps	1. Open the opportunity 2. Click "Cancel Opportunity" and confirm
Expected Result	The status changes to cancelled. Both volunteers receive a notification that names the opportunity.
Actual Result	(To be filled after test execution)
Status	(Pass/Fail/Blocked)
Priority	High

TC-028

Test Case ID	TC-028
Related Requirement(s)	FR-007
Title	View Registrations for an Opportunity
Preconditions	Logged in as a Coordinator. An opportunity has registrations.
Test Steps	1. Select the opportunity 2. Open its registration list
Expected Result	The registered volunteers and the number of available places show. No data changes.
Actual Result	(To be filled after test execution)
Status	(Pass/Fail/Blocked)
Priority	Medium

TC-029

Test Case ID	TC-029
Related Requirement(s)	FR-007
Title	Record Attendance and Hours - Valid
Preconditions	Logged in as a Coordinator. The event date has passed. Volunteers are registered.
Test Steps	1. Open the registration list 2. Mark one volunteer as attended 3. Enter 3 hours 4. Click "Save"
Expected Result	Attendance and hours save. They appear in the volunteer's participation history.
Actual Result	(To be filled after test execution)
Status	(Pass/Fail/Blocked)
Priority	Medium

TC-030

Test Case ID	TC-030
Related Requirement(s)	FR-007, NFR-006
Title	Record Hours - Invalid Hours
Preconditions	Logged in as a Coordinator. The event date has passed.
Test Steps	1. Open the registration list 2. Mark a volunteer as attended 3. Enter -2, or a letter, or leave empty 4. Click "Save"
Expected Result	The system asks for a correction. Nothing is saved.
Actual Result	(To be filled after test execution)
Status	(Pass/Fail/Blocked)
Priority	Medium

TC-031

Test Case ID	TC-031
Related Requirement(s)	FR-008
Title	Grant Coordinator Access
Preconditions	Logged in as an Administrator. A volunteer account exists.
Test Steps	1. Open the user management page 2. Select the volunteer 3. Grant coordinator access and save
Expected Result	The change saves. After the next login, the user sees coordinator features.
Actual Result	(To be filled after test execution)
Status	(Pass/Fail/Blocked)
Priority	Medium

TC-032

Test Case ID	TC-032
Related Requirement(s)	FR-008
Title	Remove Coordinator Access
Preconditions	Logged in as an Administrator. A coordinator account exists.
Test Steps	1. Open the user management page 2. Select the coordinator 3. Remove coordinator access and save
Expected Result	The change saves. The user no longer sees coordinator features.
Actual Result	(To be filled after test execution)
Status	(Pass/Fail/Blocked)
Priority	Medium

TC-033

Test Case ID	TC-033
Related Requirement(s)	FR-008
Title	Review Program Summaries
Preconditions	Logged in as an Administrator. Opportunities, registrations, and hours exist.
Test Steps	1. Open the Program Overview page
Expected Result	The summary shows counts of opportunities, registrations, attendance, and total hours. No data changes.
Actual Result	(To be filled after test execution)
Status	(Pass/Fail/Blocked)
Priority	Medium

TC-034

Test Case ID	TC-034
Related Requirement(s)	FR-008
Title	Program Summaries - No Data For Period
Preconditions	Logged in as an Administrator. No data exists for the selected period.
Test Steps	1. Open Program Overview 2. Select a period with no activity
Expected Result	A clear "no information" message shows.
Actual Result	(To be filled after test execution)
Status	(Pass/Fail/Blocked)
Priority	Low

TC-035

Test Case ID	TC-035
Related Requirement(s)	FR-009
Title	Notification - Opportunity Time or Location Changed
Preconditions	A volunteer is registered. Logged in as the Coordinator.
Test Steps	1. Edit the opportunity time or location 2. Save 3. Log in as the registered volunteer and open notifications
Expected Result	The volunteer has a notification that names the opportunity and states what changed.
Actual Result	(To be filled after test execution)
Status	(Pass/Fail/Blocked)
Priority	Medium

TC-036

Test Case ID	TC-036
Related Requirement(s)	FR-009
Title	Notification - Unregistered Volunteer Not Notified
Preconditions	A second volunteer is not registered for the opportunity.
Test Steps	1. Edit the opportunity date and save 2. Log in as the unregistered volunteer
Expected Result	The unregistered volunteer receives no notification.
Actual Result	(To be filled after test execution)
Status	(Pass/Fail/Blocked)
Priority	Low

TC-037

Test Case ID	TC-037
Related Requirement(s)	NFR-001
Title	Performance - Lists Load Within Two Seconds
Preconditions	Database holds hundreds of opportunities and thousands of records.
Test Steps	1. Open Browse Opportunities 2. Open My Volunteering 3. Measure the time to display each page
Expected Result	Each page displays within two seconds.
Actual Result	(To be filled after test execution)
Status	(Pass/Fail/Blocked)
Priority	Medium

TC-038

Test Case ID	TC-038
Related Requirement(s)	NFR-003
Title	Reliability - No Duplicate Records On Rapid Double Click
Preconditions	Logged in as a Volunteer. The opportunity has open spaces.
Test Steps	1. Double-click "Register" quickly 2. Open My Volunteering
Expected Result	Only one registration exists. Remaining spaces fall by one only.
Actual Result	(To be filled after test execution)
Status	(Pass/Fail/Blocked)
Priority	High

7. Code Repository and Branching Strategy

Repository: https://github.com/anwar7akim/volunteer-platform

- **main:** stable code. The team merges to main only at a tagged release.
- **development:** ongoing development. All feature branches merge here through a pull request.
- **Feature branches:** one branch for each key feature, for example `feature/auth`, `feature/opportunities`, `feature/registration`, `feature/coordinator-tools`. Each branch includes the unit tests for its feature.
- **Commits:** short messages in the Conventional Commits style, for example `feat(auth): add role-based login`.
- **Release tag:** the team tags the submitted version `ITR2.1`.

8. Task Allocation and Timeline

8.1 Task Breakdown

Team member	Coding	Testing	Documentation
Anwar Bonunagui	Project setup, data storage layer, role-based login (US1)	Unit tests TC-001 to TC-007, TC-038	Repository and branching section, test results
Mohammad Jalili	Browse and search (US2), register for opportunity (US3)	Unit tests TC-008 to TC-016	Traceability matrix
Jana Hussien	Post, edit, and cancel opportunities (US6), attendance and hours (US7)	Unit tests TC-023 to TC-030	Prototype overview, updated use cases
Crystal Mumtaz-Shah	Withdraw (US4), My Volunteering (US5), admin screens (US8), notifications (US9)	Unit tests TC-017 to TC-022, TC-031 to TC-037	Test plan, sprint plan updates

The team must confirm this allocation. Each member also reviews one pull request from another member.

8.2 Sprint Plan

Week	Plan
Week 1	Set up the Java project, the data storage layer, and the test framework. Implement login and role checks, browse and search, registration, and opportunity posting. Write unit tests for each.
Week 2	Implement withdraw, My Volunteering, attendance and hours, admin screens, and notifications. Refine Week 1 features. Run all test cases. Record Actual Result and Status. Update the coverage status.
Week 3	Fix defects found in testing. Finalize the prototype. Gather feedback from the instructor and teammates. Tag `ITR2.1` and submit the PDF to eClass.

9. Submission Checklist

- Push all code and documents to GitHub.
- Tag the version `ITR2.1`.
- Submit this PDF on eClass: prototype overview, updated requirements and use cases, test plan and results, updated iteration plan.